

CELTIC SEA BUOY 62023, Ireland

WMO: 996640

Lat: 51.400N

Lon: 7.920W

Elev: 0

StdP: 101.33

Time Zone: -1.00 (W01)

Period: 02-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.2	4.1	N/A	N/A	N/A	N/A	N/A	N/A	26.0	7.8	23.4	8.4	10.7	20	0.921

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.3	21.2	N/A	19.8	N/A	18.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5.4	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	19.5	17.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	8.7	8.1	8.6	10.1	12.1	14.5	16.2	16.6	15.8	13.3	10.8	9.1
	DBStd	3.53	1.73	1.69	1.80	1.76	1.66	2.00	1.61	1.31	1.72	1.78	1.89	1.99
	HDD10.0	219	46	54	48	17	1	0	0	0	0	1	13	38
	HDD18.3	2313	299	286	302	246	194	116	70	55	78	156	227	285
	CDD10.0	957	5	2	4	21	65	135	192	206	175	104	37	11
	CDD18.3	9	0	0	0	0	0	1	3	3	2	0	0	0
	CDH23.3	2	0	0	0	0	0	0	0	1	1	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	9.2	11.2	10.0	8.9	8.7	8.4	7.9	7.8	8.0	8.4	9.6	10.5	11.3
Precipitation	PrecAvg	1030	107	83	69	70	70	71	77	80	75	113	103	108
	PrecMax	1340	182	194	177	131	157	200	196	248	192	197	204	343
	PrecMin	764	33	15	14	13	2	12	19	8	12	28	28	29
	PrecStd	134	46	56	37	36	35	44	40	56	44	51	49	63
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	12.1	13.6	17.1	19.0	21.4	22.6	22.7	22.9	17.8	14.7	13.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	11.7	11.2	11.6	14.5	16.7	19.4	20.6	20.5	21.0	16.9	14.0	12.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	11.3	10.7	11.1	12.8	15.3	18.1	19.2	19.2	19.1	16.2	13.5	11.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	10.9	10.2	10.7	11.9	14.2	17.1	18.2	18.5	17.8	15.6	13.1	11.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	2.5	2.6	2.9	3.1	3.2	3.1	3.2	3.3	3.0	2.6	2.5	2.3
		MCDBR	2.7	2.5	3.4	4.3	5.1	4.8	4.5	4.5	4.9	3.1	2.6	2.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.344	0.353	0.379	0.382	0.394	0.377	0.384	0.366	0.365	0.358	0.337	0.335	
	taud	2.305	2.363	2.324	2.359	2.370	2.454	2.440	2.503	2.494	2.448	2.372	2.278	
	Ebn at Noon	648	763	812	857	861	879	866	866	825	751	657	587	
	Edh at Noon	67	83	103	111	115	107	106	95	86	75	62	59	
All-Sky Solar Radiation	RadAvg	0.80	1.53	2.70	4.44	5.40	5.96	5.40	4.58	3.33	1.89	1.02	0.62	
	RadStd	0.07	0.12	0.28	0.30	0.36	0.47	0.43	0.26	0.25	0.17	0.07	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page